

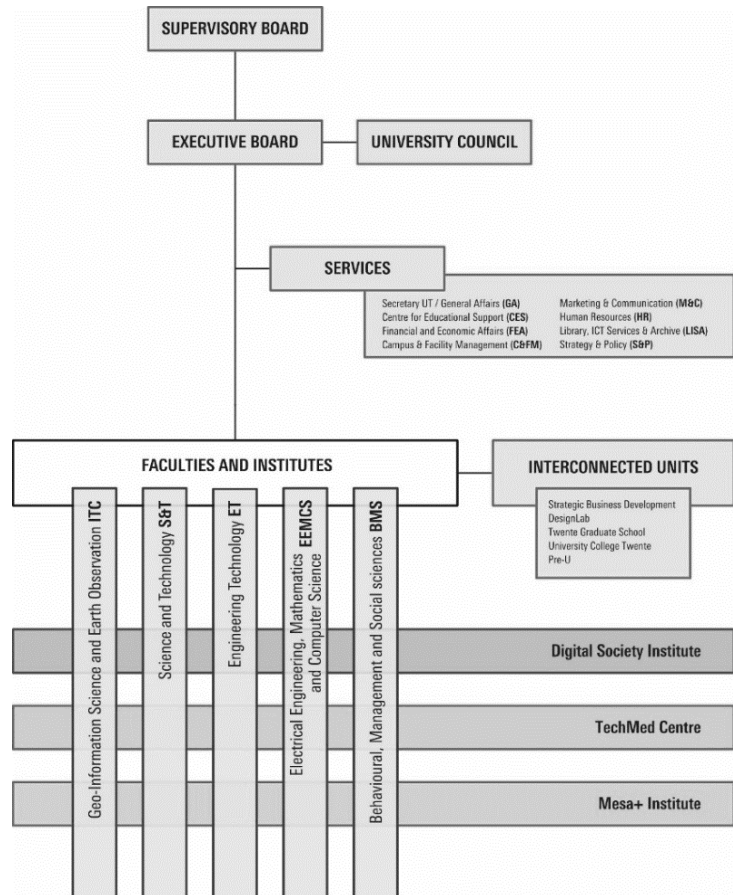
Literature Search

Academic Research Skills (Premaster)

September 2026

Ir. L. van Ewijk (Luuk)
Information Specialist
University Library

Who is this guy?



Information specialist at
the University Library

Information Literacy

Visit

Information Literacy Guides

> 1. Information Literacy

After studying this section, you will know what Information Literacy is, how these skills benefit you, and how you can master them. Among other things, you will learn who to contact for all your questions about the topic, and ho...

> 2. Types of Information

After studying this section, you will be able to distinguish different types of sources that you can use to inform yourself or to use in your own work. Among other things, you will learn how to recognize a primary source, wh...

> 3. Evaluating Information

After studying this section, you will be ready to critically assess the relevance and quality of sources. Among other things, you will learn why it is important to know where information comes from and what statistical...

> 4. Accessing Sources

After studying this section, you will be able to access all the literature needed for your assignment or research project. Among other things you will learn why some content is behind a paywall, how to easily access ebooks...

> 5. Using Information

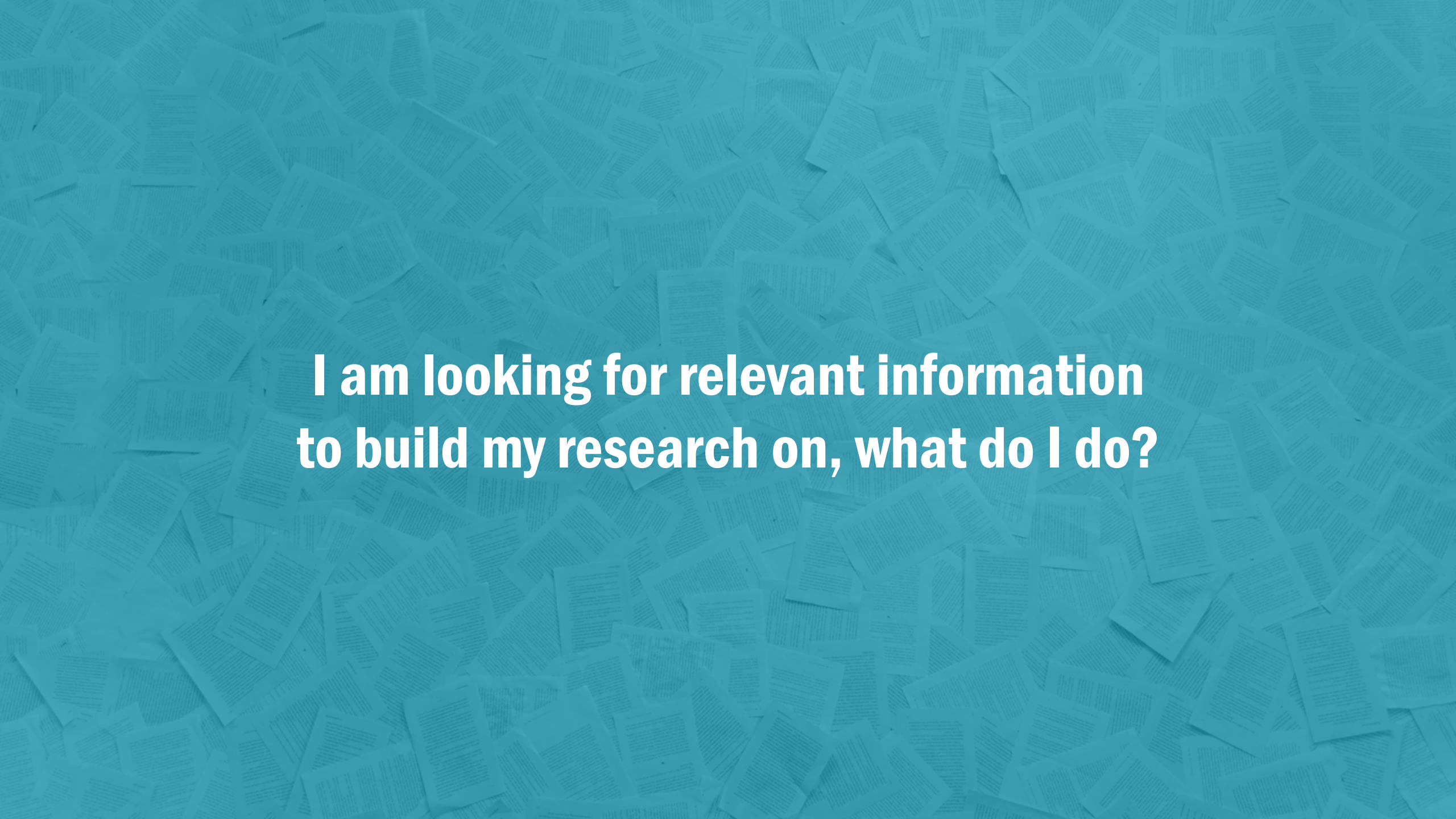
After studying this section, you will be aware of the conditions under which you can incorporate information from other sources in your own work. Among other things, you will learn how to avoid plagiarism, how to save...

> 6. Tools for Searching

After studying this section, you will understand the fundamental differences between various tools developed for literature searching. Among other things, you will learn which databases help you quickly find peer-reviewed...

> 7. Systematic Searching

After studying this section, you will understand how you can locate the most relevant academic and grey literature on your topic. Among other things, you will learn about the importance of preparing a search, about the...

The background of the image is a dense, overlapping field of small, rectangular pieces of paper, resembling newspaper clippings or index cards. These pieces are scattered across the entire frame, creating a textured, chaotic appearance. The color of the background is a solid, muted teal or blue.

**I am looking for relevant information
to build my research on, what do I do?**

Today's lecture

Before you start

Nasty questions and reference managers

Define search strategy

Starting points for a structured search process

Gather search terms

Building a vocabulary to search with effectively

Conduct searches

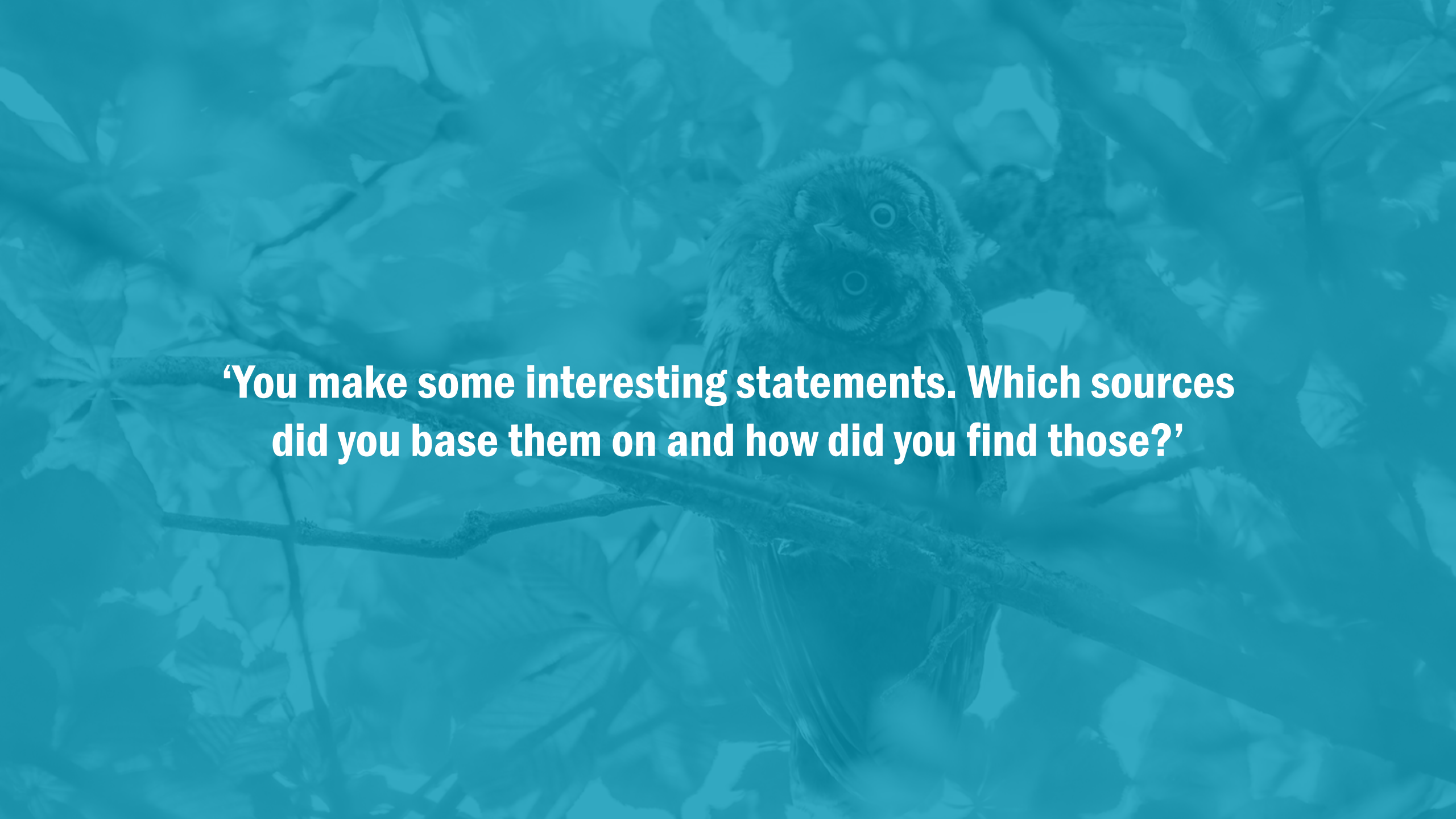
Employing various search tools and techniques



● **Before you start**







‘You make some interesting statements. Which sources did you base them on and how did you find those?’

Choose your answer

‘I googled 15 papers on my topic.’

‘Based on research question A, I performed a search for peer-reviewed literature published in English in the past 10 years in multi-disciplinary abstract database Scopus, using these search terms and this final query, finding 11 relevant papers, adding 4 through backward reference searching.’

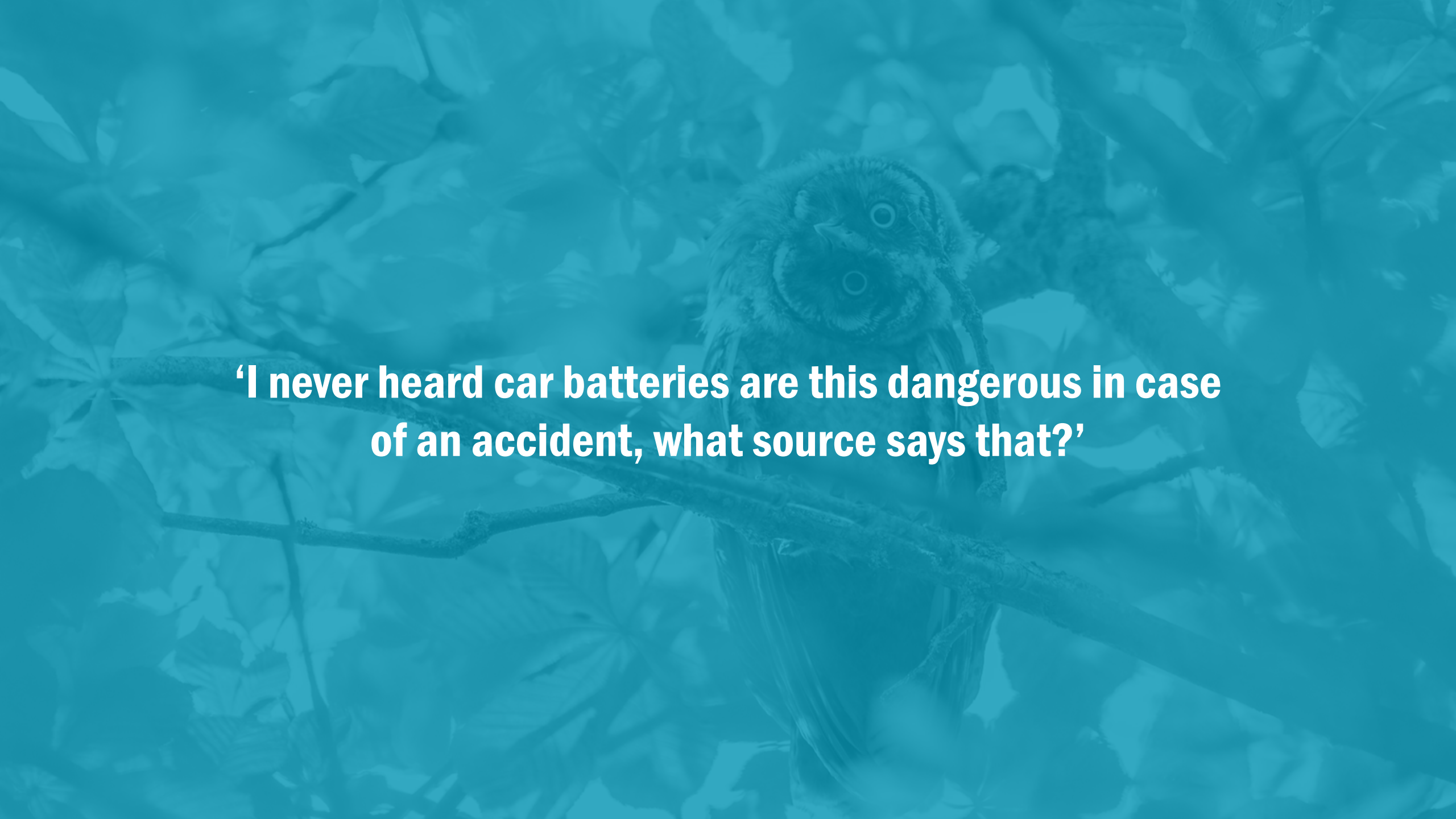


‘I think there’s been some recent developments in car battery design, did you find all research on it?’

Choose your answer

‘Let’s Google car batteries again.’

‘To make sure I include all published articles, including conference papers, on batteries for electric cars, I will do a search in the publisher database IEEE Xplore, as this database contains many journals on engineering topics and allows me to do a full-text search that I cannot perform in Scopus.’

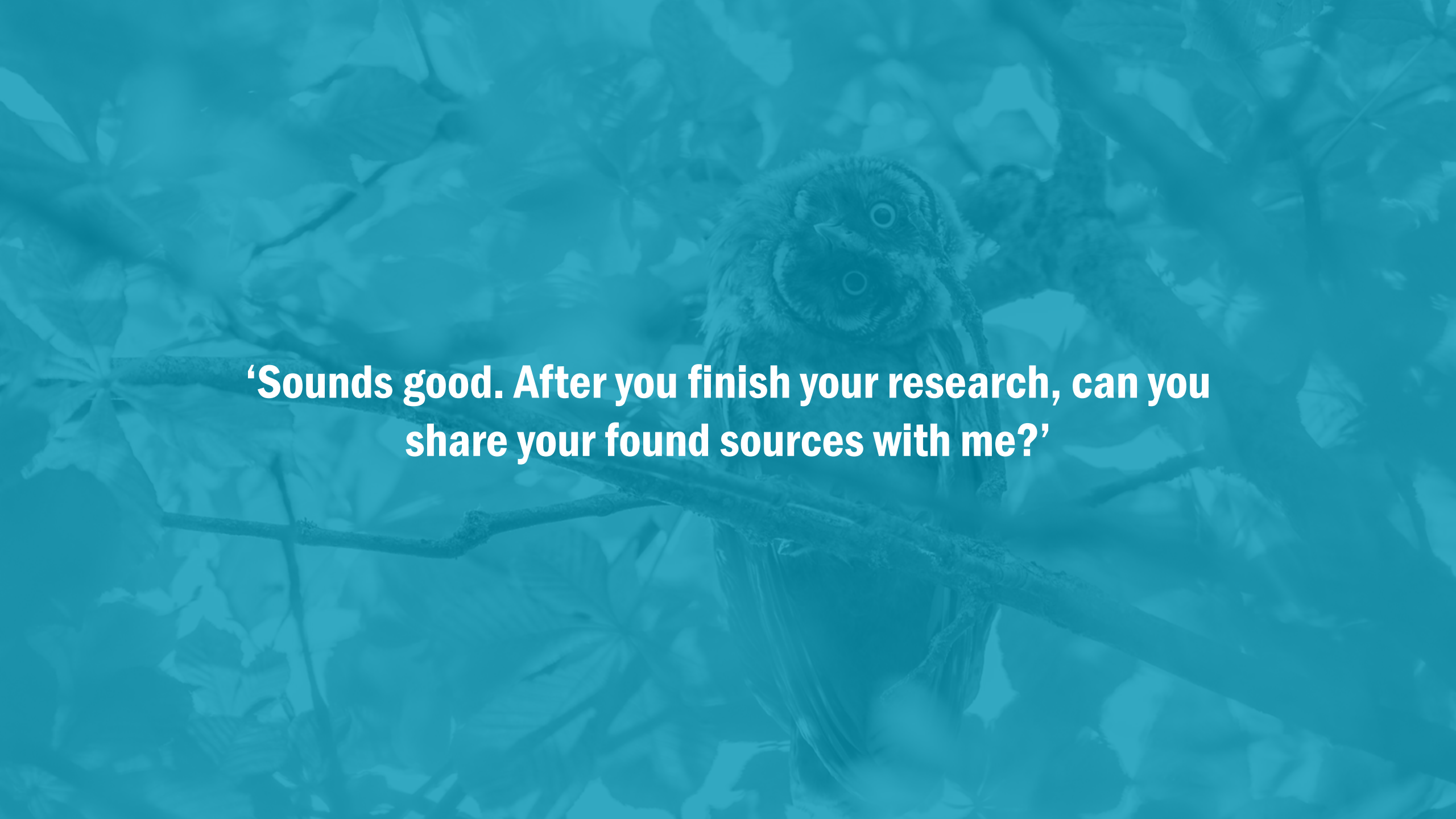


‘I never heard car batteries are this dangerous in case of an accident, what source says that?’

Choose your answer

‘A paper, so it’s
very trustworthy.’

‘Since Scopus only indexes peer-reviewed literature, I assume that works I found are written by experts in the field. I do however check if the paper is actually relevant with regard to my topic, and if its publishing date suggests I am reading about the most recent research findings. If I have any doubts about bias, I check the acknowledgement section for funder names’



‘Sounds good. After you finish your research, can you share your found sources with me?’

Choose your answer

‘Do you use
Google Drive?’

‘Definitely! I saved all my found sources in my reference manager Zotero, and I can create a group library, invite you to it and share the sources I have currently saved in my personal library named *Car Battery Research* with you.’



Well done!

Manage references

The screenshot displays the Zotero desktop application interface. The top menu bar includes 'File', 'Edit', 'View', 'Tools', and 'Help'. The left sidebar shows a tree view of the library structure, with 'Information Literacy' selected. The main pane shows a list of references with columns for 'Title' and 'Creator'. A context menu is open over the selected item, 'The Scoring Rubric for Information Literacy as a Tool for Learning', listing various management actions. The right pane shows the details for the selected item, including metadata like 'Item Type', 'Title', 'Author', 'Editor', 'Book Title', 'Series', 'Series Number', 'Volume', '# of Volumes', 'Edition', 'Place', 'Publisher', 'Date', 'Pages', 'Language', 'ISBN', 'Short Title', and 'URL'.

Library Structure (Left Sidebar):

- My Library
 - Information Design
 - Information Literacy
 - My Publications
 - Duplicate Items
 - Unfiled Items
 - Trash
- Group Libraries
 - EIS-IS

Reference List (Main Pane):

Title	Creator
> Development of a shared item repository for progress testing in veterinary education	Schaper et al.
Digital literacy as part of a new curriculum for the Netherlands	Fisser and Strijker
> Low literacy and written drug information: information-seeking, leaflet evaluation and ...	van Beusekom et...
Predictors of teachers' use of ICT in school – the relevance of school characteristics, te...	Drossel et al.
The Scoring Rubric for Information Literacy as a Tool for Learning	Van Helvoort and...

Context Menu (Over Selected Item):

- Show in Library
- Add Note
- Add Attachment
- Find Full Text
- Add to Collection
- Remove Item from Collection...
- Create Book from Book Section
- Duplicate Item
- Move Item to Trash...
- Export Item...
- Create Bibliography from Item...
- Generate Report from Item...

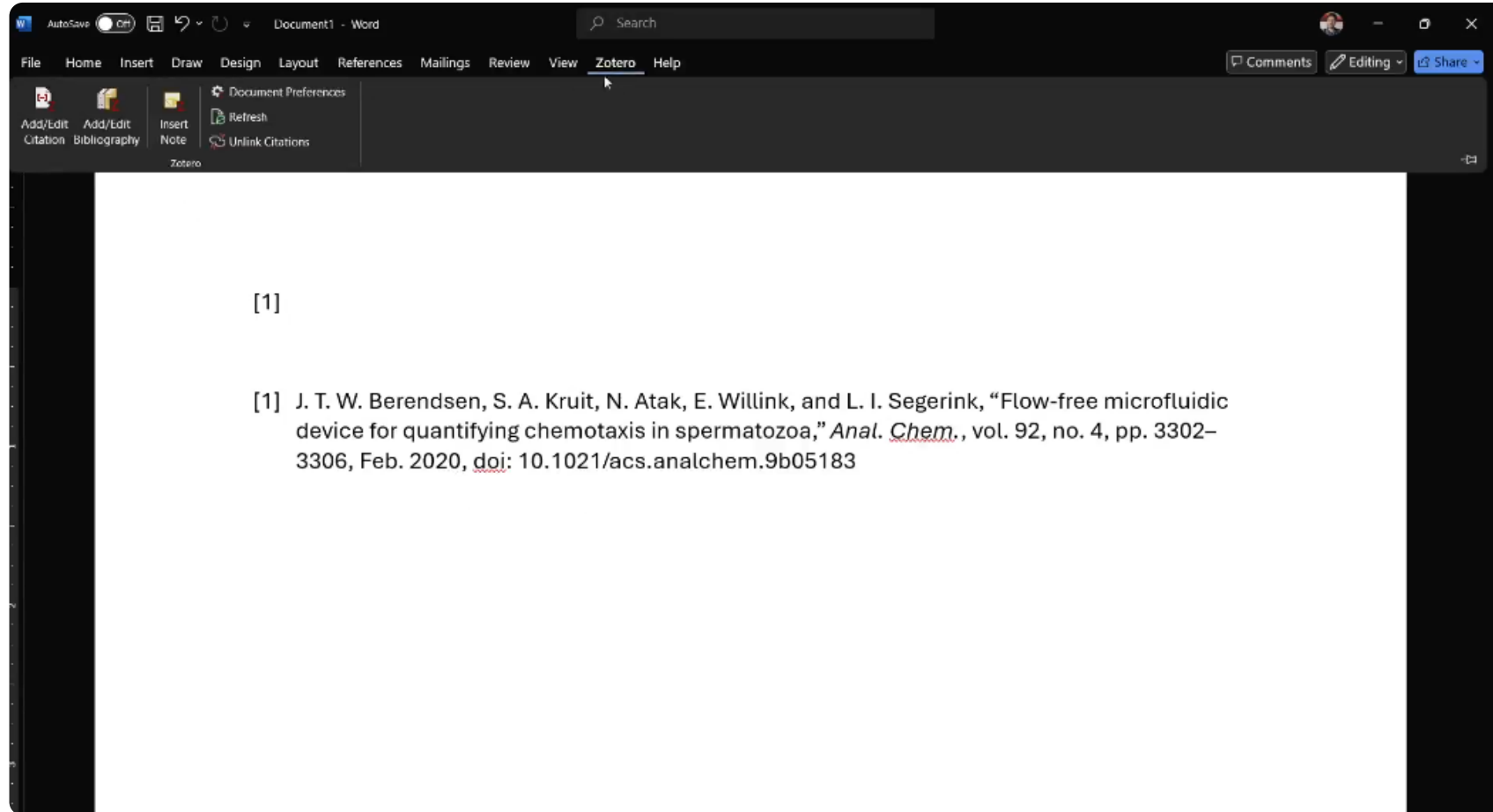
Item Details (Right Pane):

The Scoring Rubric for Information Literacy as a Tool for Learning

Info

- Item Type: Book Section
- Title: The Scoring Rubric for Information Literacy as a Tool for Learning
- Author: Van Helvoort, Jos
- Author: Joosten, Henriëtte
- Editor: Sales, Dora
- Editor: Pin..., Maria
- Book Title: Pathways into Information Literacy and Communities of Practice: Teaching Approaches and Case Studies
- Series
- Series Number
- Volume
- # of Volumes
- Edition
- Place
- Publisher: Elsevier Inc.
- Date: 2016
- Pages: 115-126
- Language: English
- ISBN: 978-0-08-100680-1
- Short Title
- URL

Manage references



Manage references

1

Visit www.zotero.org, install Zotero 10 & Zotero Connector and create an account

In case you already have an account, log in and add your UT mail address.

2

Go to *edit > settings > general > locate* and select the University of Twente

3

Go to *edit > settings > account > sync > data syncing* and add your account

Manage references

4

Login at zotero.org and go to *groups > create a new group (private membership)*

5

After creating the group go to *member settings > send more invitations*

6

Add the UT mail address of group member(s) and click on *invite members*

- **Define search strategy**

Define a search question

You will work with a set of questions relevant to your general problem description.

Define a search question

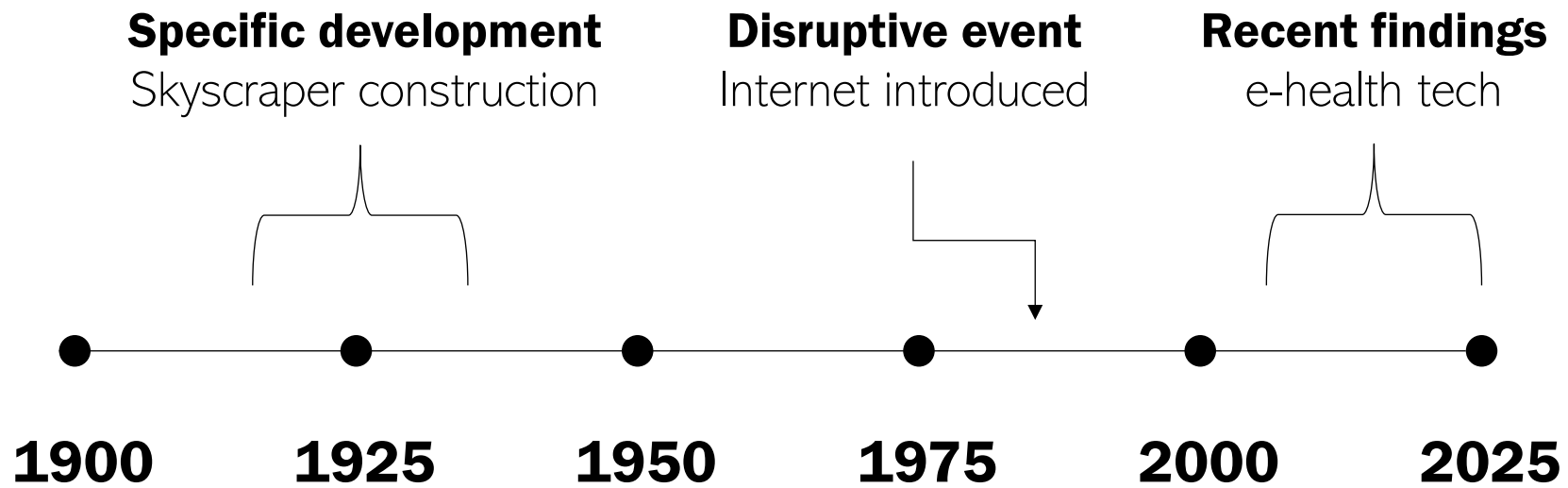
Research question

What correlation is there between the charging time of electric vehicles and the range they have?

Search question

What research has been done on the relation between the charging time and range of electric vehicles?

Determine time span



Pick languages

For 2020 publications registered with Crossref digital object identifiers (DOIs), over 85% were in English, followed by German (2.9%), Portuguese (2.2%), Spanish (2.0%) and Bahasa Indonesian (1.3%). These numbers are weakened by not including information on articles with other registration agencies, particularly relevant for Russian, Chinese and other East Asian languages in countries with their own registration agencies.

- UNESCO (2023)

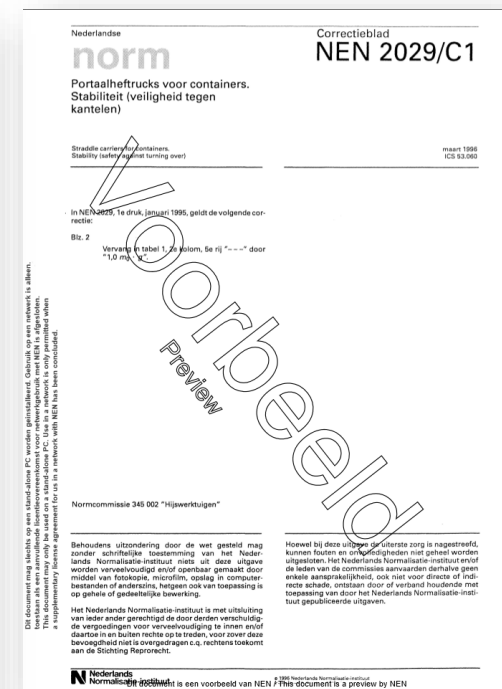
Pick languages



Academic Literature
Usually English will suffice



Grey literature
Documents in other languages might be relevant



Choose types of sources

Academic literature

Published by academic publishers

- Journal articles
- Conference papers
- Chapters of edited volumes

Usually in the form of original research or literature reviews

Grey literature

Not published by academic publishers

- Policies
- Standards
- Patents
- News articles
- Technical reports
- White papers

Select search tools

1

Visit www.utwente.nl/library, go to *Databases > Scopus*

2

Visit www.utwente.nl/library, go to *Databases > Databases A-Z > IEEE Xplore*

3

Visit Google Scholar via <https://scholar.google.com>

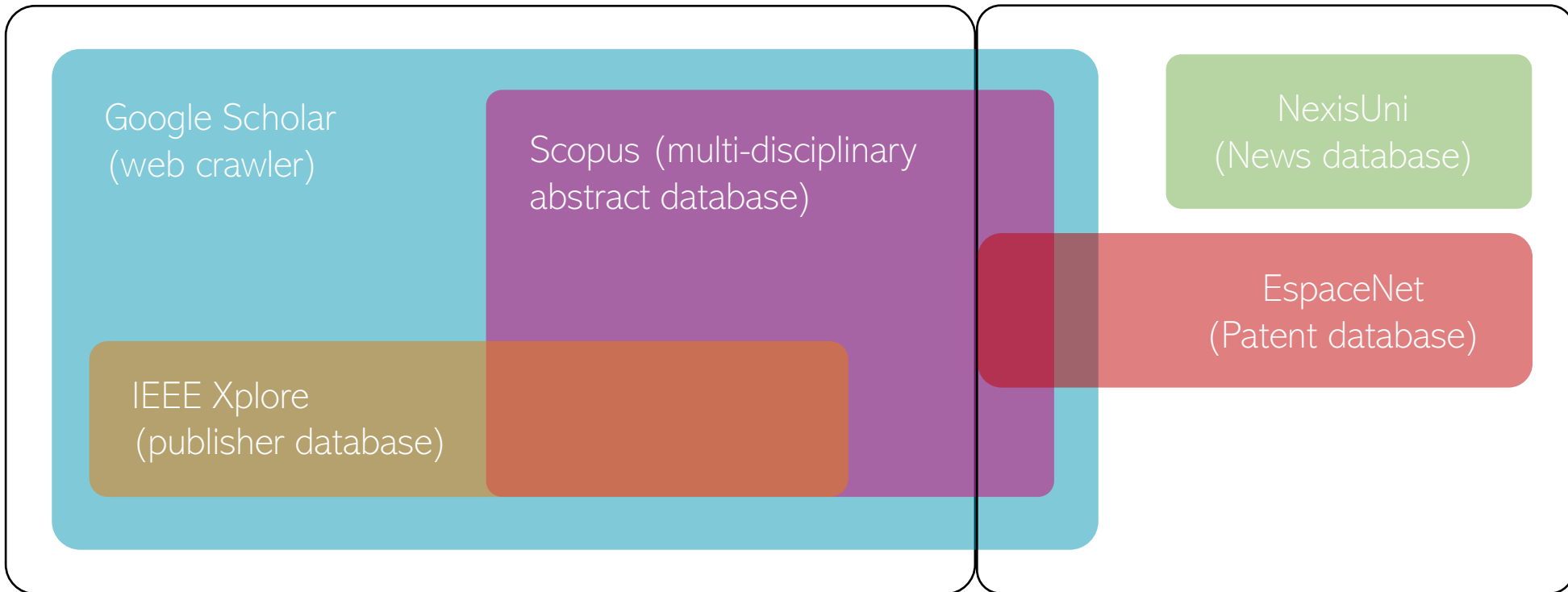
4

In all search tools, type in the word 'engineering' and click on *search*

Select search tools

Academic literature

Grey literature



* This is a simplification for educational purposes

Select search tools

Type of keyword-based search tool	Examples of these search tools	Affiliated publishers & scope content	Peer-reviewed content	Non peer-reviewed content	Limited to curated content	Allows for search in full-text
Multi-disciplinary abstract database	Scopus, Web of Science	Almost all publishers, scope multi-disciplinary	Yes	No	Yes	No
Publisher database / publisher platform	IEEE Xplore, ScienceDirect	One publisher, scope related its journals	Yes	Yes	Yes	Yes
Web crawler / search engine	Google Scholar, Semantic Scholar	Any publisher, scope multi-disciplinary	Yes	Yes	No	No

Select search tools

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- **Gather search terms**

Gather search terms

Research question

What correlation is there between the charging time of electric vehicles and the range they have?

Search question

What research has been done on the relation between the charging time and range of electric vehicles?

Gather search terms

Key Concepts	Synonyms	Narrower Terms	Broader Terms
Electric vehicles	Electric car, plug-in electric vehicles	Tesla, battery, transmission	Mobility, the grid, transport
Charging time	Charging duration	(charging) station, kW	Travel time
Range	Distance, reach	Kilometers, miles	Specs, specifications
Relation	Correlation, link	-	-

The central ideas of your topic, that will help you find the core set of literature.
Use in search: electric vehicles AND charging time AND range AND relation

Gather search terms

Key Concepts	Synonyms	Narrower Terms	Broader Terms
Electric vehicles	Electric car, plug-in electric vehicles	Tesla, battery, transmission	Mobility, the grid, transport
Charging time	Charging duration	(charging) station, kW	Travel time
Range	Distance, reach	Kilometers, miles	Specs, specifications
Relation	Correlation, link	-	-

Capture sources that use different terminology to describe the same concept.
Use in search: range OR distance OR reach

Gather search terms

Key Concepts	Synonyms	Narrower Terms	Broader Terms
Electric vehicles	Electric car, plug-in electric vehicles	Tesla, battery, transmission	Mobility, the grid, transport
Charging time	Charging duration	(charging) station, kW	Travel time
Range	Distance, reach	Kilometers, miles	Specs, specifications
Relation	Correlation, link	-	-

For example, zoom in on aspects of a key concept to refine your scope.

Use in search: electric vehicle AND (battery OR transmission) AND range

Gather search terms

Key Concepts	Synonyms	Narrower Terms	Broader Terms
Electric vehicles	Electric car, plug-in electric vehicles	Tesla, battery, transmission	Mobility, the grid, transport
Charging time	Charging duration	(charging) station, kW	Travel time
Range	Distance, reach	Kilometers, miles	Specs, specifications
Relation	Correlation, link	-	-

For example, expand your search to include relevant but less specific sources.
Use in search: charging time AND (mobility OR transport) AND range



● Conduct searches

Access sources

Open Access literature

Access for everyone
(with internet connection)

Closed access literature

Access through
University Library

Access sources

- Through library catalogue FindUT you can check if we have access
- Scopus, Web of Science and Google Scholar link to FindUT
- The Lean Library browser extension will tell you if you have access
- On campus you always have access

Closed access literature



Access through
University Library

Access sources

1

Visit www.utwente.nl/library, and search for 'engineering' in FindUT

2

In FindUT, try the different filter options under *Held By Library*

3

In Scopus, click on an 'availability' button in the list of search results

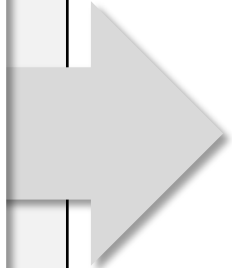
4

In Google Scholar, add UT library links via *settings > library links*

Access sources

Open Access literature

Closed access literature

- A version uploaded in a repository might be findable via Google Scholar
 - Ask the corresponding author to share it (for individual use)
 - Request via *Interlibrary Loan*
- 

Conduct searches

1

Find core set of relevant papers

Keyword-based search in Scopus



Export list to Zotero via Connector

2

Find additional relevant papers

Item-based (reference) search in Scopus



Export individual papers to Zotero via Connector

3

Find full-text PDFs of all papers

Find through Zotero or download via Google Scholar



Read and annotate PDFs of papers in Zotero

1. Find core set of papers

Title

Keywords

Abstract

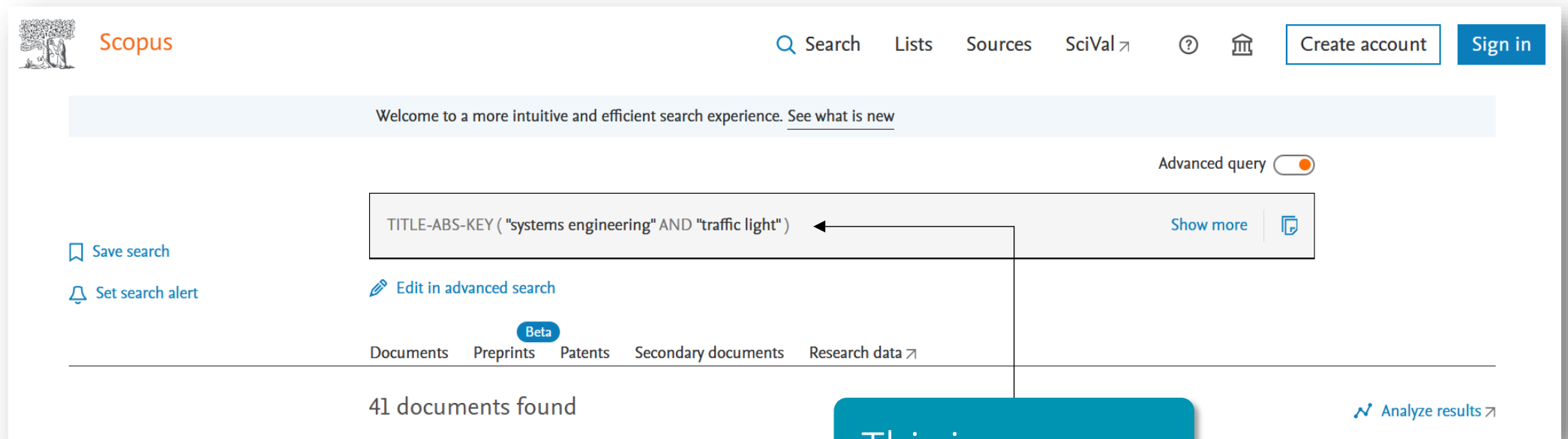
Full-text



1. Find core set of papers

Key Concepts	Synonyms	Narrower Terms	Broader Terms
Electric vehicles	Electric car, plug-in electric vehicles	Tesla, battery, transmission	Mobility, the grid, transport
Charging time	Charging duration	(charging) station, kW	Travel time
Range	Distance, reach	Kilometers, miles	Specs, specifications
Relation	Correlation, link	-	-

1. Find core set of papers



The screenshot displays the Scopus search interface. At the top left is the Scopus logo. The top navigation bar includes links for Search, Lists, Sources, SciVal, a help icon, a library icon, and buttons for 'Create account' and 'Sign in'. A welcome message states: 'Welcome to a more intuitive and efficient search experience. [See what is new](#)'. Below this, there is a search bar containing the query 'TITLE-ABS-KEY ("systems engineering" AND "traffic light")'. To the right of the search bar is a 'Show more' link and a document icon. Above the search bar is an 'Advanced query' toggle switch, which is currently turned on. To the left of the search bar are links for 'Save search' and 'Set search alert'. Below the search bar is a link for 'Edit in advanced search'. A horizontal menu below the search bar includes 'Documents', 'Preprints', 'Patents' (marked with a 'Beta' badge), 'Secondary documents', and 'Research data'. Below this menu, it states '41 documents found'. In the bottom right corner, there is a link for 'Analyze results'. A blue callout box with the text 'This is a query or search string' has an arrow pointing to the search query in the search bar.

Scopus

Search Lists Sources SciVal ? ? Create account Sign in

Welcome to a more intuitive and efficient search experience. [See what is new](#)

Advanced query ☒

TITLE-ABS-KEY ("systems engineering" AND "traffic light") Show more

Save search Set search alert Edit in advanced search

Documents Preprints Patents Beta Secondary documents Research data

41 documents found Analyze results

This is a query or search string

1. Find core set of papers

These filters will be reflected in your query

The screenshot displays a research database interface. On the left, a 'Refine results' sidebar contains various filters: Access type, Year, Author name, Subject area, Document type, Publication stage, Source title, Keyword, Affiliation, Funding sponsor, Country/territory, Source type, and Language. Each filter has a dropdown arrow. A blue callout box points to the 'Access type' filter, stating 'These filters will be reflected in your query'. The main area shows a list of search results with columns for Document title, Authors, Year, and Source. The results are sorted by 'Cited by (highest)', as indicated by the 'Sort on:' dropdown menu on the right. A blue callout box points to the 'Relevance' option in the sort menu, stating 'Use 'relevance' as sorting mechanism!'. The search results list includes:

	Document title	Authors	Year	Source	
☐ 1	Cratering and blowing soil by rocket engines during lunar landing	Metzger, P.T., Lane, J.E.	2010	Lunar Settlements, pp. 551-576	
	View abstract				
☐ 2	Star 37S solid rocket motor thermostructural analysis			Studies in Applied Mechanics 35(C), pp. 405-422	3
	View abstract	Availability	Related documents		
☐ 3	Optimal robust design of hybrid rocket engines	Pastrone, D., Casalino, L.	2016	Springer Optimization and Its Applications 114, pp. 269-285	2
	View abstract	Availability	Related documents		
☐ 4	Integrated design-trajectory optimization for hybrid rocket motors	Pastrone, D., Casalino, L.	2013	Springer Optimization and Its Applications 73, pp. 343-362	2
	View abstract	Availability	Related documents		
☐ 5	An Introduction to Flow-Induced Instabilities in Rocket Engines	d'Agostino, L., Cervone, A., et al.	2017	CISM International Centre for Mechanical Sciences, Trento, Italy	1

1. Find core set of papers

TITLE ("electric vehicles" AND (distance OR range) AND NOT agricultur*)

Field code
(part of the
record you
search in)

Exact phrase
(together as
one term)

Boolean operator
(must contain)

Boolean operator
(contain either/both)

Boolean operator
(must exclude)

Truncation
(stem of word
+ variations)

1. Find core set of papers

1

In Scopus, enter a query on your topic via *Advanced document search*

2

Sort on *relevance* and click on Zotero Connector (folder icon) in the taskbar

3

Select the first 10 papers and click on OK, in Zotero click the refresh button

2. Find additional papers

The screenshot shows the Scopus interface for a paper titled "Boundaries of high-power charging for long-range". The page includes a header with the Scopus logo, search bar, and navigation links. The paper title is partially obscured by two teal callout boxes. The left box, labeled "Forward reference search (literature that refers to this paper)", points to the "Cited by (26)" tab. The right box, labeled "Backward reference search (literature this paper is based on)", points to the "References (26)" tab. The "References (26)" tab is currently selected, showing a list of references. On the right side, there are metrics: 26 citations (62th percentile) and an FWCI of 0.72. The bottom of the page shows a filter section with "All" and "Export" options.

Scopus

Search Sources SciVal ? ? Create account Sign in

Back

Boundaries of high-power charging for long-range

Forward reference search (literature that refers to this paper)

Backward reference search (literature this paper is based on)

View PDF Full text ▾ Export ▾ Save to list

Document Impact Cited by (26) References (26) Similar documents

You can view, sort, and filter all of the References in [search results format](#).

☐ All ▾ Export

2. Find additional papers

- 1 In Scopus, click on the first search result, then on the tab 'references'
- 2 Click on a reference which title seems relevant to your search question
- 3 Click on Zotero Connector (paper icon) in the taskbar, refresh Zotero

3. Find full-text PDFs

Title	Creator	
> [icon] Development of a shared item repository for progress testing in veterinary education	Schaper et al.	[icon]
[icon] Digital literacy as part of a new curriculum for the Netherlands	Fisser and Strijker	
> [icon] Low literacy and written drug information: information-seeking, leaflet evaluation and ...	van Beusekom et...	[icon]
[icon] Predictors of teachers' use of ICT in school – the relevance of school characteristics, te...	Drossel et al.	
[icon] The Scoring Rubric for Information Literacy as a Tool for Learning	Van Helvoort and...	

Show in Library

Add Note

Add Attachment >

Find Full Text

Add to Collection >

Remove Item from Collection...

Create Book from Book Section

Duplicate Item

Move Item to Trash...

Export Item...

Create Bibliography from Item...

Generate Report from Item...

Find & download PDF via
Google Scholar / FindUT

Only shows for references
with DOIs (digital object
identifiers)

3. Find full-text PDFs

1

In Zotero, check if 'find full-text' is available for references without a PDF

2

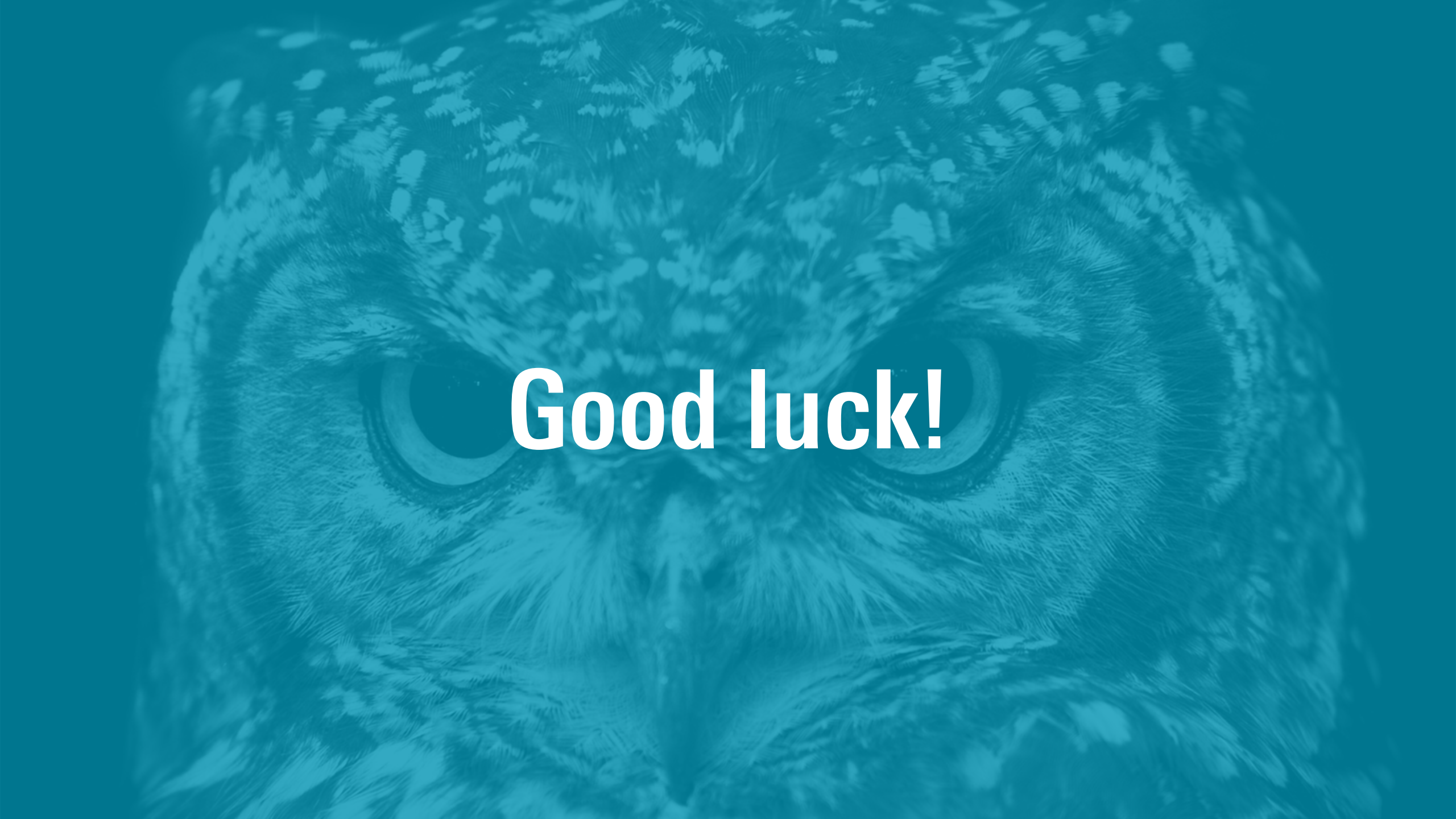
For those left, search in Google Scholar and click on 'library links'

3

Download the PDFs and add them to the references via 'add attachment'





A close-up, monochromatic teal image of an owl's face, serving as the background for the text.

Good luck!